# Experiment 3: Creating DataFrames & Analyzing Properties

**PIN:** 241UDS0096

**Aim:** To create Pandas DataFrames using NumPy arrays and CSV files, and analyze DataFrame properties.

## Step 1: Import libraries

If NumPy or Pandas is not installed, run `%pip install numpy pandas` in a separate cell.

In [1]:
```
# %pip install numpy pandas
import numpy as np
import pandas as pd
```

## Step 2: Create a 2-D NumPy array

In [2]:
```
# Four weeks of sales data for three branches
sales_241UDS0096 = np.array([
    [245, 380, 415, 520],
    [310, 275, 490, 365],
    [420, 535, 330, 610]
])

print(sales_241UDS0096)
print('Shape:', sales_241UDS0096.shape)
print('Dimensions:', sales_241UDS0096.ndim)
```

```
[[245 380 415 520]
 [310 275 490 365]
 [420 535 330 610]]
Shape: (3, 4)
Dimensions: 2
```

## Step 3: Convert the array into a DataFrame

In [3]:
```
df_index_241UDS0096 = ['Branch_A', 'Branch_B', 'Branch_C']
df_columns_241UDS0096 = ['Week_1', 'Week_2', 'Week_3', 'Week_4']

df_241UDS0096 = pd.DataFrame(
    data=sales_241UDS0096,
    index=df_index_241UDS0096,
    columns=df_columns_241UDS0096
)
display(df_241UDS0096)
```

| | Week_1 | Week_2 | Week_3 | Week_4 |
|---|---|---|---|---|
| **Branch_A** | 245 | 380 | 415 | 520 |
| **Branch_B** | 310 | 275 | 490 | 365 |
| **Branch_C** | 420 | 535 | 330 | 610 |

## Step 4: Add a new column

In [4]:
```
df_241UDS0096['Week_5'] = [290, 440, 375]
display(df_241UDS0096)
```

| | Week_1 | Week_2 | Week_3 | Week_4 | Week_5 |
|---|---|---|---|---|---|
| **Branch_A** | 245 | 380 | 415 | 520 | 290 |
| **Branch_B** | 310 | 275 | 490 | 365 | 440 |
| **Branch_C** | 420 | 535 | 330 | 610 | 375 |

## Step 5: Add a new row

In [5]:
```
branch_d_241UDS0096 = pd.DataFrame({
    'Week_1': [350],
    'Week_2': [465],
    'Week_3': [390],
    'Week_4': [575],
    'Week_5': [425]
}, index=['Branch_D'])

df_241UDS0096 = pd.concat([df_241UDS0096, branch_d_241UDS0096])
display(df_241UDS0096)
```

| | Week_1 | Week_2 | Week_3 | Week_4 | Week_5 |
|---|---|---|---|---|---|
| **Branch_A** | 245 | 380 | 415 | 520 | 290 |
| **Branch_B** | 310 | 275 | 490 | 365 | 440 |
| **Branch_C** | 420 | 535 | 330 | 610 | 375 |
| **Branch_D** | 350 | 465 | 390 | 575 | 425 |

## Step 6: Read the CSV file

In [7]:
```
stu_df_241UDS0096 = pd.read_csv('retail_inventory_241UDS0096.csv')
display(stu_df_241UDS0096)
```

| | product | category | branch | units_sold | unit_price | stock_remaining |
|---|---|---|---|---|---|---|
| **0** | Notebook | Stationery | Kakinada | 145 | 55 | 80 |
| **1** | Pen Pack | Stationery | Rajahmundry | 230 | 30 | 150 |
| **2** | Water Bottle | Accessories | Kakinada | 86 | 220 | 34 |
| **3** | Backpack | Bags | Amalapuram | 54 | 850 | 22 |
| **4** | Pencil Box | Stationery | Rajahmundry | 112 | 75 | 65 |
| **5** | Geometry Kit | Stationery | Kakinada | 73 | 140 | 41 |
| **6** | Highlighter Set | Stationery | Amalapuram | 68 | 95 | 29 |
| **7** | File Folder | Stationery | Kakinada | 91 | 45 | 58 |
| **8** | Calculator | Electronics | Rajahmundry | 37 | 650 | 16 |
| **9** | Sticky Notes | Stationery | Amalapuram | 104 | 40 | 70 |
| **10** | Marker Set | Stationery | Kakinada | 59 | 110 | 27 |
| **11** | Ruler | Stationery | Rajahmundry | 125 | 25 | 90 |

## Step 7: Analyze DataFrame properties

In [8]:
```
# First five rows
display(stu_df_241UDS0096.head())

# Shape
print('Shape:', stu_df_241UDS0096.shape)

# Index
print('Index:', stu_df_241UDS0096.index)

# Data types
print(stu_df_241UDS0096.dtypes)

# Column names
print(stu_df_241UDS0096.columns)
```

| | product | category | branch | units_sold | unit_price | stock_remaining |
|---|---|---|---|---|---|---|
| **0** | Notebook | Stationery | Kakinada | 145 | 55 | 80 |
| **1** | Pen Pack | Stationery | Rajahmundry | 230 | 30 | 150 |
| **2** | Water Bottle | Accessories | Kakinada | 86 | 220 | 34 |
| **3** | Backpack | Bags | Amalapuram | 54 | 850 | 22 |
| **4** | Pencil Box | Stationery | Rajahmundry | 112 | 75 | 65 |

```
Shape: (12, 6)
Index: RangeIndex(start=0, stop=12, step=1)
product            object
category           object
branch             object
units_sold          int64
unit_price          int64
stock_remaining     int64
dtype: object
Index(['product', 'category', 'branch', 'units_sold', 'unit_price',
       'stock_remaining'],
      dtype='object')
```

## Step 8: DataFrame information

In [10]:
```
stu_df_241UDS0096.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 6 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   product          12 non-null     object
 1   category         12 non-null     object
 2   branch           12 non-null     object
 3   units_sold       12 non-null     int64 
 4   unit_price       12 non-null     int64 
 5   stock_remaining  12 non-null     int64 
dtypes: int64(3), object(3)
memory usage: 708.0+ bytes
```

## Step 9: Descriptive statistics

In [11]:
```
display(stu_df_241UDS0096.describe())
```

| | units_sold | unit_price | stock_remaining |
|---|---|---|---|
| **count** | 12.000000 | 12.000000 | 12.000000 |
| **mean** | 98.666667 | 194.583333 | 56.833333 |
| **std** | 51.772814 | 268.594476 | 37.990031 |
| **min** | 37.000000 | 25.000000 | 16.000000 |
| **25%** | 65.750000 | 43.750000 | 28.500000 |
| **50%** | 88.500000 | 85.000000 | 49.500000 |
| **75%** | 115.250000 | 160.000000 | 72.500000 |
| **max** | 230.000000 | 850.000000 | 150.000000 |

## Step 10: Transpose rows and columns

In [12]:
```
display(stu_df_241UDS0096.head(10).transpose())
```

Columns 1–8 of 11

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| **product** | Notebook | Pen Pack | Water Bottle | Backpack | Pencil Box | Geometry Kit | Highlighter Set |
| **category** | Stationery | Stationery | Accessories | Bags | Stationery | Stationery | Stationery |
| **branch** | Kakinada | Rajahmundry | Kakinada | Amalapuram | Rajahmundry | Kakinada | Amalapuram |
| **units_sold** | 145 | 230 | 86 | 54 | 112 | 73 | 68 |
| **unit_price** | 55 | 30 | 220 | 850 | 75 | 140 | 95 |
| **stock_remaining** | 80 | 150 | 34 | 22 | 65 | 41 | 29 |

Columns 9–11 of 11 · first column repeated

| | 7 | 8 | 9 |
|---|---|---|---|
| **product** | File Folder | Calculator | Sticky Notes |
| **category** | Stationery | Electronics | Stationery |
| **branch** | Kakinada | Rajahmundry | Amalapuram |
| **units_sold** | 91 | 37 | 104 |
| **unit_price** | 45 | 650 | 40 |
| **stock_remaining** | 58 | 16 | 70 |

## Result

DataFrames were created using a NumPy array and a CSV file. Properties including shape, index, column names, data types, information, descriptive statistics, adding a row and column, and transposition were analyzed.